

Task 1:

OSI 7-Layer Model Diagram and Examples

Objective: Represent the OSI 7-layer reference model with correct numbering, layer names, and associated protocols/hardware devices.

1. Text-Based OSI 7-Layer Model Diagram:

Layer 7: Application Layer	Example: HTTP, HTTPS, FTP, SMTP
Layer 6: Presentation Layer	Example: SSL/TLS, JPEG, ASCII
Layer 5: Session Layer	Example: NetBIOS, PPTP, RPC
Layer 4: Transport Layer	Example: TCP, UDP protocols
Layer 3: Network Layer	Example: Router, IP Address
Layer 2: Data Link Layer	Example: Network Switch, MAC Address
Layer 1: Physical Layer	Example: Ethernet Cable (Cat6), Hub

2. Layer Details & Real-World Examples:

- **Layer 7 - Application Layer:** Provides application-level interface for network access. (*Examples: HTTP, HTTPS, Web Browsers*)
- **Layer 6 - Presentation Layer:** Translates, encrypts, and compresses data formats. (*Examples: SSL/TLS Encryption, JPEG/PNG*)
- **Layer 5 - Session Layer:** Establishes, manages, and terminates network sessions. (*Examples: NetBIOS, RPC*)
- **Layer 4 - Transport Layer:** Manages end-to-end data transfer, segmentation, and reliability. (*Examples: TCP, UDP*)
- **Layer 3 - Network Layer:** Handles logical IP addressing, packet creation, and routing across networks. (*Examples: Router, IP Protocol*)

- **Layer 2 - Data Link Layer:** Handles physical MAC addressing, frame creation, and error detection on local link. *(Examples: Network Switch, NIC)*
- **Layer 1 - Physical Layer:** Transmits raw binary bits over physical hardware mediums. *(Examples: Ethernet Cable Cat6, Network Hub, Fiber Optics)*

Task 2:

Matching Networking Scenarios to OSI Layers

Objective: Correctly map specific networking tasks and functions to their corresponding OSI model layers.

- **a) Encrypting data before sending:** Presentation Layer (Layer 6)
- **b) Routing packets across networks:** Network Layer (Layer 3)
- **c) Converting data to electrical signals:** Physical Layer (Layer 1)
- **d) Opening a web page in your browser:** Application Layer (Layer 7)
- **e) Detecting and correcting errors in frames:** Data Link Layer (Layer 2)

Task 3:

Application Data Flow Across OSI Layers (WhatsApp Example)

Objective: Explain how at least three OSI layers are involved in transmitting data for a daily mobile application (WhatsApp).

Selected App: WhatsApp

1. Application Layer (Layer 7):

- **Role:** Acts as the user interface where the user types and sends a text message or image. It uses application protocols like HTTP/HTTPS and WebSockets to initiate the data transfer request.

2. Presentation Layer (Layer 6):

- **Role:** Responsible for data formatting and security. Before the message leaves the device, this layer applies **End-to-End Encryption (SSL/TLS / Signal Protocol)** to secure the

message content and compresses media files (photos/videos) for efficient transmission.

3. Network Layer (Layer 3):

- **Role:** Responsible for logical addressing and routing. It encapsulates the encrypted data packets with the device's source IP address and the destination WhatsApp server IP address, enabling routers across the Internet to direct the packets to the correct target destination.

Task 4:

Troubleshooting Ping vs. Web Browsing Failure

Objective: Identify the specific OSI layer(s) responsible for a network issue where IP ICMP ping succeeds, but web browsing (HTTP/HTTPS) fails.

1. Identified OSI Layers Causing the Problem:

- **Primary Fault: Layer 7 (Application Layer)**
 - **Secondary Faults: Layer 6 (Presentation Layer) and Layer 5 (Session Layer)**
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2. Technical Reasoning & Justification:

- **Why Layers 1, 2, and 3 are Functional:**

A successful ping uses the ICMP protocol, which operates at **Layer 3 (Network Layer)**. This confirms that physical cables, network switches (Layer 2), IP routing (Layer 3), and basic connectivity are functioning properly.

- **Why the Issue is at Layer 7 (Application Layer):**

Web browsers rely on **HTTP (Port 80)** and **HTTPS (Port 443)** protocols at Layer 7. The failure to load the web page suggests that while the host IP is reachable, the Web Server software (e.g., Apache/Nginx) is down, an Application-layer firewall is blocking HTTP/HTTPS traffic, or a DNS issue is preventing hostname resolution.

- **Why the Issue could be at Layer 6 or Layer 5:**

At **Layer 6 (Presentation Layer)**, an expired or invalid SSL/TLS security certificate can block the encrypted HTTPS session from establishing. At **Layer 5 (Session Layer)**, firewall session tables might be dropping TCP handshake connections for web ports.

